walnut

Tencent Cloud AI CAN DO IT Hackathon · Singapore 2026

FINTECH TRACK · ASPIRE · THE INTERNAL BRAIN

Internal Brain

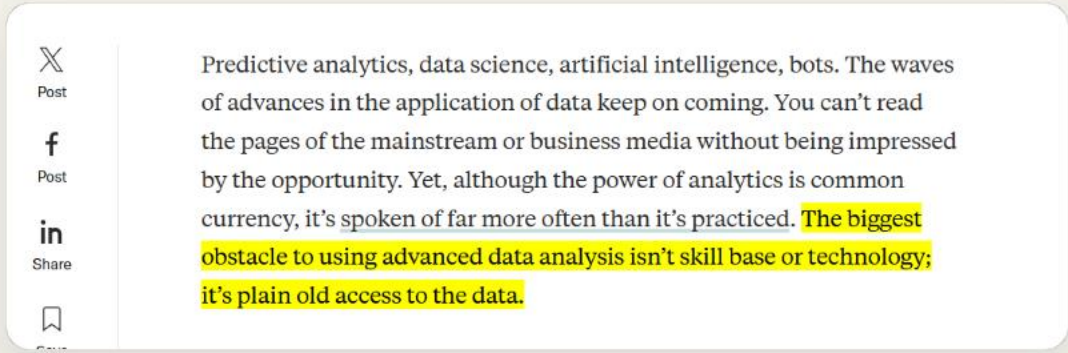
Ask one question across Confluence, Jira, Slack and Google Drive. Get a cited answer built **only from what you are allowed to open**, with every access on an audit trail nobody can quietly edit.

Permission-checked twice | Tamper-evident audit | Built with CodeBuddy · lives in WorkBuddy

The data already exists

The biggest obstacle to using advanced data analysis isn't skill base or technology; it's plain old access to the data.

SOURCE · HARVARD BUSINESS REVIEW



02 · THE COST

Searching takes a quarter of the week

The answers are somewhere in the organisation. Getting to them is where the time goes.

25%

of a team's time goes to searching for answers.

ATLASSIAN · STATE OF TEAMS 2025 · 12,000 KNOWLEDGE WORKERS

5.3 hours

a week, per person, waiting on colleagues for information or recreating work that already exists.

PANOPTO WITH YOUGOV · 2018 · 1,001 EMPLOYEES

42%

of an organisation's know-how is held only by individual employees. When they move on, it goes with them.

PANOPTO WITH YOUGOV · 2018

The cost shows up as slower projects, repeated work and decisions made on partial information.

03 · THE RISK

AI projects stall before they start

Lack of AI-ready data puts AI projects at risk.

60%

of AI projects unsupported by AI-ready data will be abandoned through 2026, Gartner forecasts.

SOURCE · GARTNER NEWSROOM · FEB 2025



04 · THE FINTECH EDGE

In a fintech, a leak can be a crime

An assistant that can read everything can also leak everything. For a payments company supervised by MAS, that blocks rollout entirely.

01
Tipping-off
Telling a customer that a suspicious-transaction report exists is an offence (CDSA s.57; MAS Notice PSN01).

02
Card data
Card numbers pasted into a ticket must never reach an index or a model (PCI DSS 3.4.1).

03
Oversharing and injection
Assistants oversharing, and instructions hidden in retrieved documents (EchoLeak, Slack AI), are on record.

04
Stale permissions
Synced permissions lag the source by 30 minutes to a day. A removed person keeps seeing what they should not.

So the answer has to be useful and permission-safe at the same time.

The model must never see anything the person asking cannot see, and compliance must be able to prove it afterwards.

05 · WHAT WE HEARD

Teams told us where the hours go

01

Months before a project can start

Each new data or AI project spends six to eight months gathering data from scattered systems. The next project starts the hunt again.

02

Access that arrives late

Data approvals drag on, so interns, new joiners and project teams sit waiting for the data they need.

03

Tools chosen by old systems

Data held inside one vendor's ecosystem keeps teams on that vendor's tools, even when better ones exist.

04

Knowledge that leaves with people

When a senior person moves on, the handover note holds only part of what they knew.

ASPIRE · HACKATHON KICK-OFF

"What can we offer this customer?" is the question that "takes time right now".

ASPIRE'S HEAD OF AI

OUR INTERVIEWS

Teams in aviation, asset management, IT services, insurance, sovereign wealth and health tech confirmed the same bottleneck.

WHAT WE DO

We make any organisation's scattered data immediately usable: connected, understood and trusted, and only ever shown to the people allowed to see it.

THE SOLUTION

What is our solution?

Two layers on one foundation: a **Company Brain** that connects your data, and the **Internal Brain** that answers each person only from what they are allowed to open, with every step on the record.

SOLUTION · TODAY

Company knowledge lives in a dozen places

Each team keeps its own records, in its own systems and formats, each with its own permission rules.

WORK TOOLS

- Confluence pages
- Jira issues
- Slack threads
- Google Drive files

DATA PLATFORMS

- Databases (SQL)
- Data-lake files (Parquet, Arrow, CSV)
- Warehouse exports
- Spreadsheets

DOCUMENTS

- PDFs, incl. scanned
- Word and PowerPoint
- Email and mailboxes
- Policies and runbooks

PEOPLE

- Meeting notes
- Handover notes
- Who can open what
- Contractors and guests

Every system decides who may see what in its own way.

Confluence page restrictions, Jira project roles and issue security, Slack private and Connect channels, Drive inheritance. An assistant has to respect all of them, live.

SOLUTION · WITH WALNUT

From scattered information to one record

Walnut connects information across your systems and brings it into one record, with every value pointing back to where it came from.

Confluence

DB Migration Plan (v6) · payment-gateway space

Jira

DBMIG-12 · Ledger cluster cutover (wave 4)

Slack

#db-migration · "Replication lag on ledger-shard-3..."



ONE CONNECTED RECORD

Database migration project

Status · Wave 4: ledger cutover planned for 24 Oct `jira DBMIG-12`

Blocker · Replication lag on ledger-shard-3 `slack #db-migration`

Plan · Postgres 11 → 16, four waves `confluence cf-1001`

Every line opens to the source it came from, checked for the person asking.

SOLUTION · HOW IT WORKS

How Walnut makes it usable and trusted

01

Connect

Reads databases, documents, spreadsheets, email and the systems your teams use.

02

Understand

Recognises the same customer, project or person across every source and links them into one record.

03

Convert

Serves that record in the shape each job needs: answers, reports, analytics tables, AI context.

04

Prove

Every figure points to the cell, line or page it came from. A value without proof is refused.

05

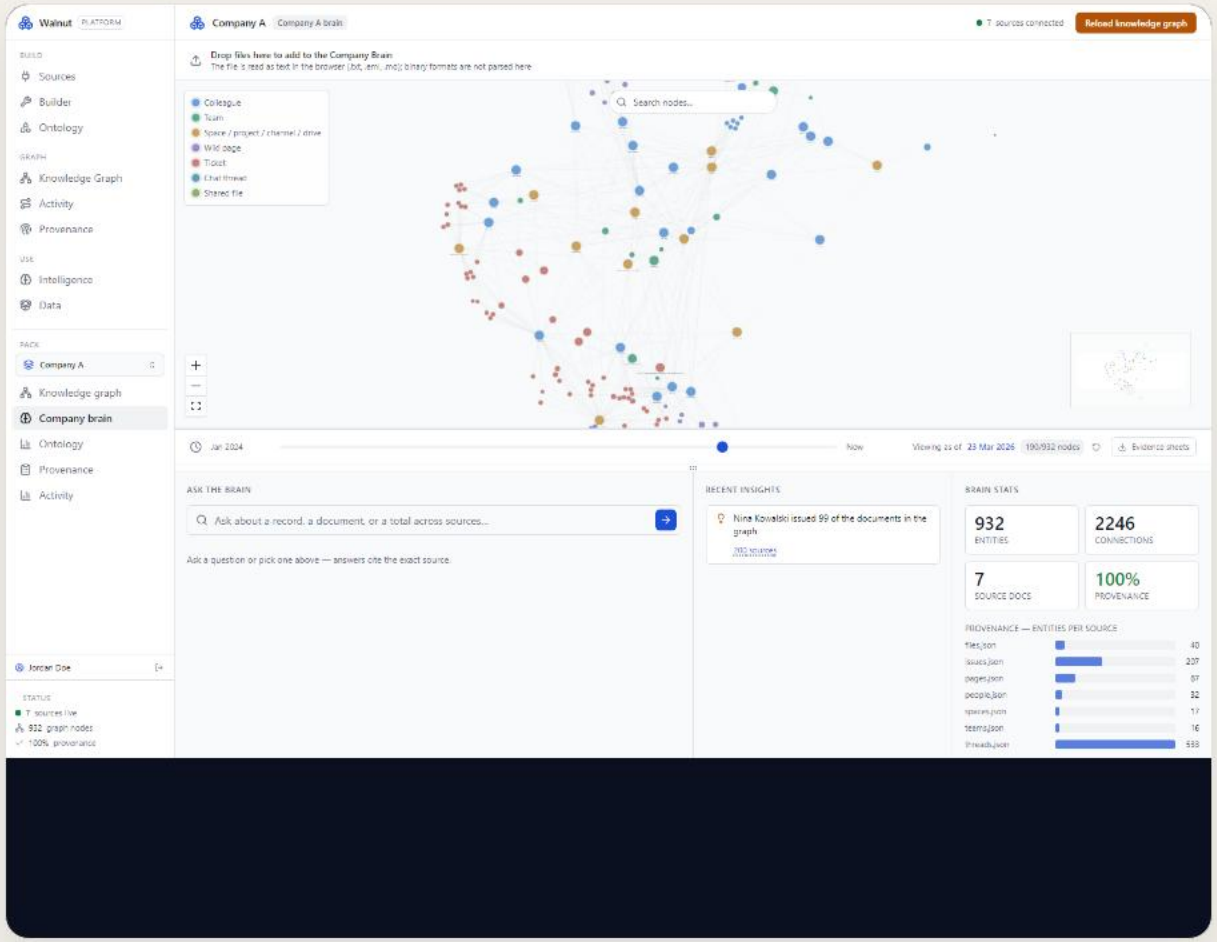
Protect

Runs inside your environment with the model you choose. Each person sees only what they may open.

SOLUTION · LAYER 1 · COMPANY BRAIN

Inside the Company Brain

Walnut's original screens, running on our synthetic fintech, Company A.



Relationship map

How records connect: page to project, ticket to team, person to space.

Full history

When each fact was true and when it was recorded. Drag the time slider to rewind the company.

Source snapshots

A write-once copy of what each source said when read, so figures stay checkable.

Change tracking

When a source value changes, every answer built on it is flagged.

SOLUTION · LAYER 2 · INTERNAL BRAIN

How one question is answered safely

Two permission checks before the model sees anything, a grounding check after it answers, and an audit record before anything is shown.

01
Who's asking
A signed token. No prompt or tool argument can name a person.

02
Check 1 · index
Search only what this person may open, with ranking statistics from those documents only.

03
Check 2 · live
Ask Confluence, Jira, Slack or Drive again, now. Error or timeout means no.

04
Answer
The model sees only passages that passed, and must cite one for every sentence.

05
Record
Written to the sealed audit log before the answer is shown. Log down means no answer.

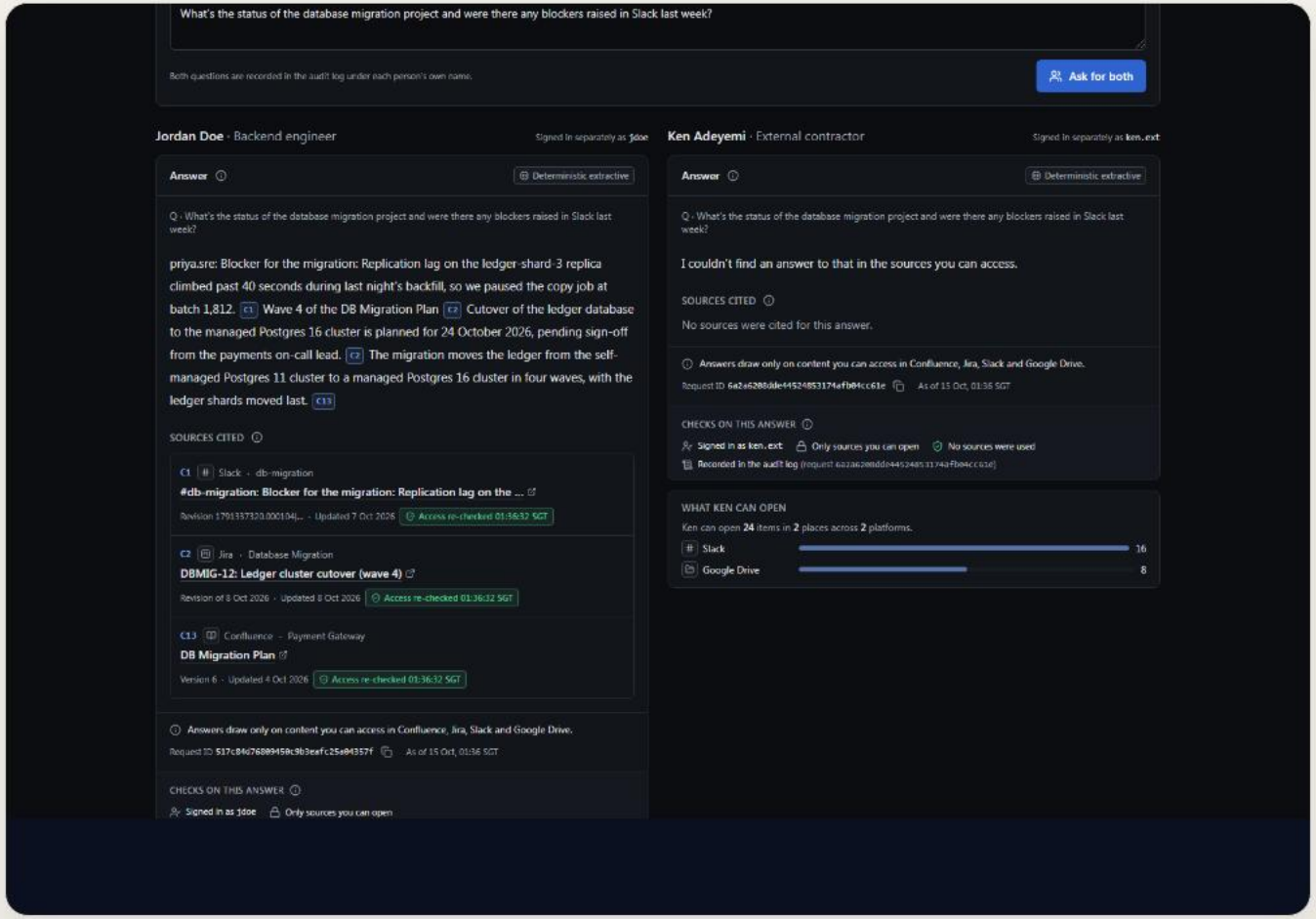
Restricted looks exactly like non-existent
The refusal is the same bytes whether the thing is hidden from you or was never there, so nothing hints that more exists.

Your model, the same checks
Tencent Hy3 through TokenHub, or any OpenAI-compatible model. The checks do not depend on the model.

IN PRACTICE

Same question, two people

Jordan, a backend engineer, and Ken, an outside contractor, ask the same thing.



1

Jordan gets the full picture

A cited answer from Slack, Jira and Confluence, every source re-checked live.

2

Ken gets a polite no

"I couldn't find an answer to that in the sources you can access." No hint more exists.

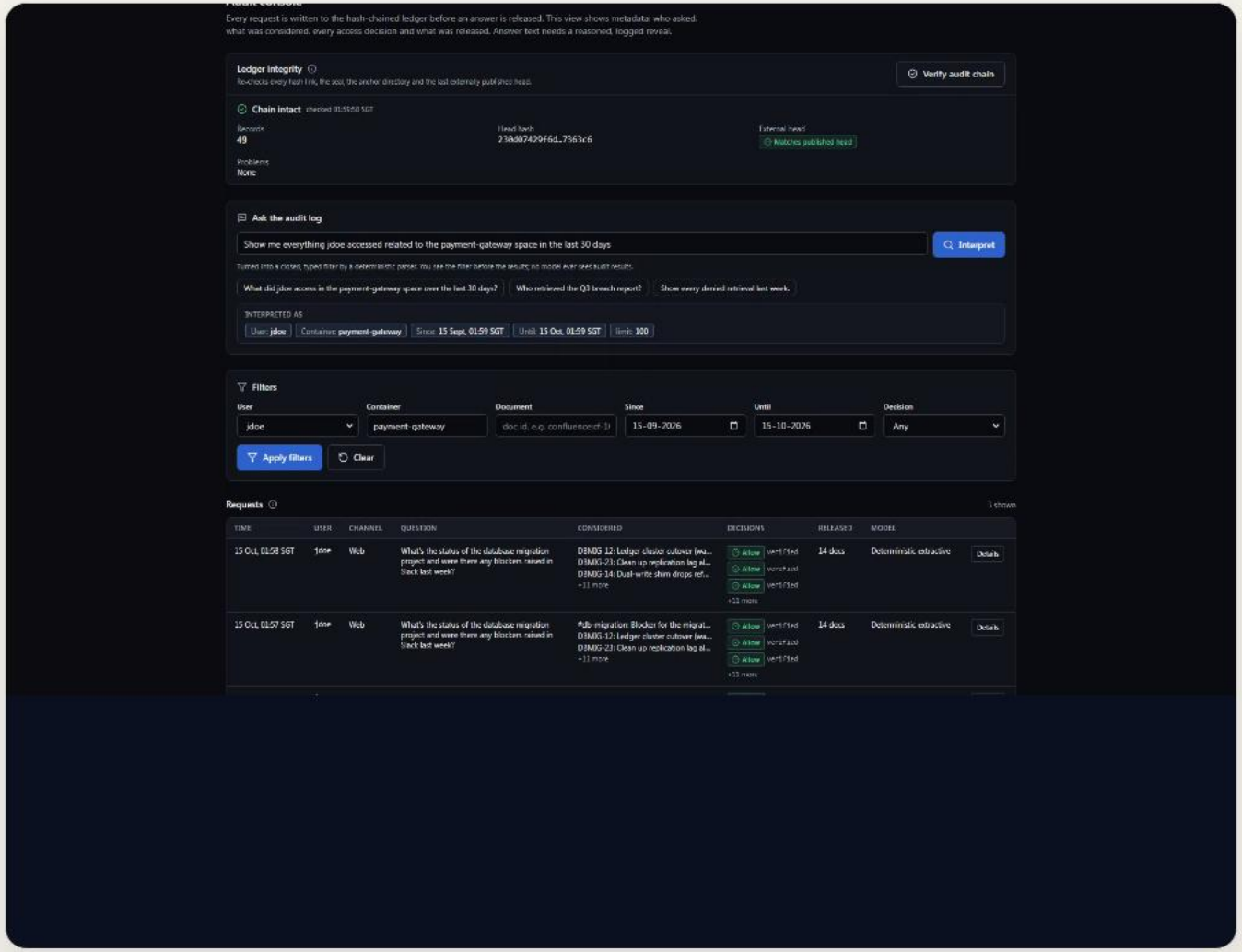
3

Both are logged

Each question, decision and answer is on the audit trail.

IN PRACTICE · COMPLIANCE

Who asked what, and proof nobody edited it



Ask the audit log in plain English

"Everything Jordan accessed in the payment-gateway space, last 30 days" becomes a visible filter.

A reason to read an answer

Compliance sees metadata by default; reading an answer needs a reason, and that is logged too.

6 / 6

tamper attacks caught by "Verify audit chain": edits, deletions, rollback, a key-holder rewrite.

PROOF

Checked against answers we already knew

We built a synthetic fintech, Company A, with 1,143 documents and 49 planted facts, so every right answer was known in advance.

0

leaks in 5,958 attacks

Restricted content in any answer, prompt or response.

LEAK EVALUATION · OCT 2026

0

differences in 400 pairs

"Restricted" and "doesn't exist" give the same answer.

DIFFERENTIAL TEST

0

injection failures

Instructions planted on all four platforms were treated as content.

976 INJECTION REQUESTS

677

automated tests

639 on the Internal Brain, 38 in the browser, all passing.

REPOSITORY TEST SUITES

The harness ships with the code, so a pilot runs the same checks on your own permissions.

USE CASES

Where a fintech gets its time back

What slows teams down today	What changes with the Internal Brain	What you get back
“What can we offer this customer?” takes time.	Approved, market-specific answers for ops and account managers, with sources.	**Faster customer answers**
Account managers must not tip off a customer.	They get the approved script, never the case notes, worded like any other answer.	**No tipping-off risk**
Access reviews take weeks of log digging.	Compliance asks the audit log in plain English, with tamper-proof records.	**Audits in minutes**
Contractors and new joiners wait for access.	Access follows each person's permissions at the source, checked live.	**Safe, faster onboarding**
Know-how leaves when people move on.	Decisions and context stay in the record with the evidence behind them.	**Knowledge that stays**

SECURITY AND CONTROL

Your data stays inside your walls

Runs in your environment

On your machines or your cloud. Embeddings are local, so nothing leaves at ingest.

Your choice of model

Tencent Hy3 via TokenHub, any OpenAI-compatible model, or none: the quoted-answer path needs no model.

Access checked twice, live

Each person sees only what the source system lets them open, re-checked at answer time.

A person stays in control

Compliance must give a reason to read an answer, and that reveal is logged.

Sealed, verifiable log

Chained and sealed, with its head published off-host. Edits and deletions are detected.

Sensitive values masked

Card numbers, CVVs, IBANs and NRICs are masked before indexing, so no model ever sees them.

Mapped to MAS TRM, PCI DSS and CDSA, with the limits stated: docs/tencent/MAS_TRM_MAPPING.md.

BUILT WITH TENCENT

Tencent tools, used for real

Every claim here has a transcript, screenshot or test in the repository.

01
CodeBuddy
Six build sessions: trust boundaries, the WorkBuddy connectors, the token tool, model-chain tests, a security review, then fixes to its own code.

02
WorkBuddy
The second front door: connectors that ask as Jordan, as Ken, and for compliance, each with its own signed token.

03
Hunyuan Hy3 · TokenHub
First in the answer chain, with DeepSeek as backup. Wired in and tested; live calls wait on account verification.

04
Lighthouse
A hardened one-command deploy kit: HTTPS, passcode-gated demo, only ports 80 and 443 open.

05
Miora
Made the cover art from our written brief.

Built on the Walnut evidence engine.

The Internal Brain adds the two permission checks, the audit console, the web app, the fintech rules and the WorkBuddy connectors on top of it.

BUSINESS VALUE

Measure it in hours saved and risk removed

TIME BACK

Hours per person per week × People × Loaded hourly cost

= Weekly cost of searching, waiting and rework

EXAMPLE · A 600-PERSON FINTECH

Saving 1.5 hours a week at US$31 an hour is about **US$1.3M a year**. Break-even for a US$100k deployment is **7 minutes a week** per person.

A projection, not a measurement. We measure it with you on one workflow, against your own baseline.

RISK REMOVED

An assistant MAS supervision can accept

Without permission-aware retrieval and an audit trail, an internal AI assistant cannot be rolled out at all.

Prepared customer conversations

Account history, approved offers and open issues, cited, before every call.

Audits that take minutes

Every access reconstructable, with proof the record was not altered.

DIFFERENTIATION

Why Walnut wins

Capability	Data warehouses	Search / RAG tools	Feature stores	Large AI platforms	Walnut
Answers only from what each person may open, checked live	no	synced ACLs	no	synced ACLs	✓
“Restricted” indistinguishable from “doesn't exist”	no	no	no	no	✓
Tamper-evident audit of every question and answer	partial	no	no	logs	✓
Connects databases, documents, chat and tickets	partial	partial	no	✓	✓
Every figure traces to its cell, line or page	lineage only	chunks	features	platform	✓
Runs in your environment with any model	cloud-bound	varies	✓	enterprise	✓

Only Walnut spans the Company Brain, live permission checks and an audit trail you can prove.

THE TEAM

BACKED BY BSF · SUTD VIE

Built by people who live the data problem



Dev

CS + BUSINESS @ NTU CCDS

AI and Automation at Singlife · AI/ML Engineer at Neurowyzr · AI and Automation at Bajaj Finserv · Researcher, NTU-Schaeffler Lab · Tech Lead, NTU AI&ML Club.



Mehul

CS @ NTU

Production AI agent over 16,000+ records at Changi's AI/ML Lab · Privacy-aware clinical AI at Neurowyzr · Multi-agent AI research at A*STAR.



Aaruni

USTEP & ASD @ SUTD

Research Intern at SEC · Architecture Intern at HighSpan, MAK Designs, TAS · Design Intern at WGS.



Ishaan

CSD @ SUTD

Real-time power-grid and industrial cybersecurity research at SUTD's iTrust Centre · Production full-stack systems at Bajaj Finance and client projects.

NEXT STEP

Start with one workflow Aspire repeats

Pick one job that costs your team hours every week. We set it up inside your environment, measure the result against your baseline, and you decide what comes next.

01
See it on your permissions
Run our leak harness against Aspire's own permission model in a sandbox.

02
Agree the workflow
Pick one: the "what can we offer this customer" question, or compliance access reviews.

03
Connect in your environment
Confluence, Jira, Slack and Drive, with Hy3 through TokenHub.

04
Measure
Hours saved and audit time against your own baseline.

05
Decide what comes next
Expand to more teams, with the full record always yours to keep.

github.com/IshaanSirbhaiya/Tencent

Start at INTERNAL_BRAIN.md · demo video in the submission